

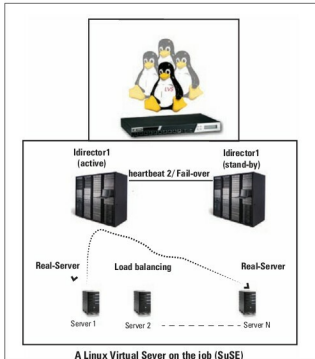


Figure 3: Linux Virtual Server

during load balancing requests: availability and scalability. *Heartbeat2* is a program that sits across load balancers, and makes sure that there is an active-standby configuration in a healthy state, so in case of a failure, the other load balancer can take over the work of the failed unit.

The *ldirector* daemon takes care of scalability, by making sure that all cluster machines are active, and in a healthy state, ready for processing.

gh availability behaviour

- Node-level monitoring:** If one of the nodes (*ldirector1*/*ldirector2*) running cluster resources stops sending out heartbeat signals, declare it dead, reboot the node and fail over all resources to a different node.
- Service-level monitoring:** If the VIP or *ldirector* service fails, try to restart the service. If this fails, reboot the node, and fail over all resources to a different node.
- Service 'stickiness':** If a dead or stand-by node becomes active again, keep the resources where they run now, and don't fail back.

Heartbeat2

It sits on both load-balancer machines, and also monitors the *ldirector* daemon.

The main high-availability configuration file (*/etc/ha.d/ha.cf*) contains the following (in our scenario):

```
crm on
udpport 694
bcast eth0
```

```
node ldirector1 ldirector2
```

/etc/ha.d/authkeys contains...

```
auth 1
1 sha1 YourSecretKey
```

This file is not readable or writeable by anyone other than the root user:

```
chmod 600 /etc/ha.d/authkeys
```

Name resolution

Add node names to */etc/hosts* on both *linux-directors*:

```
192.168.0.10 ldirector1
192.168.0.20 ldirector2
```

To start the cluster, run the following command:

```
/etc/init.d/heartbeat start
Node: ldirector2 (f8f2ad4a-a05d-416a-92a9-66b759768fb9): online
Node: ldirector1 (5792135e-ed53-438b-8a71-85f0285464c2): online
```

On the *linux-directors*, set the password for the user as: *hcluster*

```
/usr/bin/paswd hcluster
```

Configuration: linux-director (load balancer)

It is recommended that you disable *SuSEfirewall2* for the configuration, to avoid networking issues.

```
rcSuSEfirewall2 stop
chkconfig SuSEfirewall2_init off
chkconfig SuSEfirewall2_setup off
```

The *linux-directors* must be able to route traffic to the real servers. This is achieved by enabling the kernel IPv4 packet forwarding system, as follows:

```
# /etc/sysctl.conf
net.ipv4.ip_forward = 1
```

Ldirector

Create the file */etc/ha.d/ldirector.cf* and add the following code:

```
checktimeout=3
checkinterval=5
autoreload=yes
logfile="/var/log/ldirector.log"
quiescent=yes
virtual=192.168.0.200:80
fallback=127.0.0.1:80
```